StreetLightIQ: A Mobile Sensing System for City-Scale Streetlight Monitoring and Roadway Illumination Estimation

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Abstract

Public street lighting is essential for urban safety and mobility, making regular measurement of streetlight operation and estimation of useful illumination critical to urban infrastructure management. However, existing monitoring approaches often rely on manual inspections or costly infrastructure retrofits to determine streetlight functionality, without adequately evaluating their contribution to roadway and public space illumination. To address this gap, we present StreetLightIQ (SLIQ): an end-to-end edge-cloud system that leverages commodity smartphones mounted on moving vehicles for automated streetlight monitoring and illumination estimation. SLIQ addresses challenges posed by small streetlight targets, visual confounders, and heterogeneous nighttime conditions through an integrated pipeline combining a lightweight detector (trained through knowledge distillation and optimized using INT8 quantization) with temporal tracking, operational-state classification, and semantic segmentation. A physics-based model further reconstructs streetlight geometry and estimates relative roadway illumination without calibrated lux measurements, thereby obtaining per-lamp and street-level illumination metrics, including illumination coverage and failure criticality rankings. We evaluate SLIQ using nighttime drives across diverse urban road segments. The on-device detector achieves 0.78 mAP@50, compared to 0.81 for its larger teacher model, while operational-state classification and semantic segmentation attain 85.6% accuracy and 0.607 mIoU, respectively. Across 80 streets, criticality of failure rankings achieve a median Spearman correlation of 0.94, with the top-ranked lamp among the three most critical in 98% of the cases. On known lamp layouts embedded in real drives, lamp heights and positions are recovered with median errors of 0.35 m and 1.25 m, and the modelit share of each street follows the true share (Spearman correlation = 0.93, median error = 0.013). Our code is available through https://sliq-monitoring-system.github.io/streetlightiq/.

1 Introduction

Public street lighting is an essential component of urban infrastructure, playing a vital role in urban mobility, economic development, and quality of life. By illuminating roads, pedestrian pathways, and other public spaces, street lighting facilitates the safe movement of people and goods, supports commercial and social activities after dark, and enhances public security. Its significance is particularly evident in the disproportionate risks associated with nighttime travel. Nearly half of all road traffic fatalities occur after dark, despite nighttime travel accounting for only 25% of total vehicle movement [13]. Pedestrians are particularly vulnerable, with nearly 76% of pedestrian fatalities occurring after sunset [12] and injury severity nearly doubling under nighttime conditions [28]. Globally, approximately 92% of road traffic fatalities occur in low and middle income countries [34]. Beyond road safety, an updated meta-analysis found that improved street lighting was associated with a 14% reduction in crime [32]. While reliable street lighting is essential for sustaining safe and economically vibrant urban environments, its operation and maintenance at city-scale present multifaceted challenges for urban local bodies (ULB).

Street lighting networks typically comprise thousands of geographically distributed luminaires. Their operation and upkeep requires substantial recurring expenditure on electricity, maintenance, and infrastructure upgrades. Depending on the size of the city and efficiency of the system, street lighting can account for up to 60% of a city's electricity expenditure [11, 15]. Beyond energy costs, ensuring operational reliability presents additional challenges. Individual luminaires may malfunction, deteriorate over time, or provide inadequate illumination despite remaining operational. Such conditions can arise at different locations and times, making timely detection and maintenance difficult. Periodic manual inspections are time consuming and difficult to scale across extensive urban road networks. Moreover, they may miss faults arising between successive visits, thereby delaying corrective action and prolonging inadequate lighting conditions. Effective infrastructure management therefore requires accurate, up-to-date information on streetlight locations, operational status, and illumination adequacy. However, acquiring such information remains challenging for ULBs in developing countries, which often lack the financial resources and monitoring technologies available to their counterparts in developed nations. This disparity underscores the need for cost-effective, scalable, automated approaches to monitoring individual streetlights and estimating city-wide lighting conditions.

Advances in sensing and communication technologies have enabled several approaches to automated streetlight monitoring. Centralized management systems use networked controllers installed on individual luminaires to remotely monitor their operational status and faults [1]. Although effective, these systems require substantial investment in retrofitting existing luminaires and maintaining the associated communication infrastructure. Satellite-based monitoring offers large-scale observations of nighttime illumination, but lacks the spatial resolution necessary to assess individual streetlights [21]. In contrast, drive-by sensing leverages moving vehicles as mobile data collection platforms to monitor streetlights along their routes. By utilizing existing vehicular mobility rather than deploying extensive fixed monitoring infrastructure, drive-by sensing offers a promising alternative for city-wide monitoring without requiring modifications to existing luminaires [20]. However, current approaches have largely focused on detecting individual streetlights and classifying their operational status, with limited attention to evaluating the useful illumination provided to surrounding roads and public spaces.

In this paper, we present StreetLightIQ (SLIQ): an end-to-end edge-cloud system for automated streetlight monitoring at city scale. SLIQ leverages commodity smartphones deployed in existing urban mobility networks, such as radio taxis and app-based cab services.

A lightweight mobile application captures nighttime frames using the rear-facing camera of a smartphone mounted on the vehicle's dashboard or windshield. Following initial setup, the sensing and processing pipeline operates autonomously, enabling opportunistic monitoring without dedicated inspection fleets, modifications to the vehicles, or modifications to existing streetlight infrastructure. The regular movement of these vehicles offers the potential for extensive spatial coverage and repeated observations, although sustained participation may require monetary or regulatory incentives. SLIQ processes the captured observations to identify individual streetlights, classify their operational status, and estimate their useful illumination coverage. By combining commodity sensing hardware, edge-cloud processing, and existing vehicular mobility, SLIQ is designed to be deployable without additional infrastructure, to cover large road networks quickly by riding on vehicles already in circulation, and to estimate each lamp's contribution to useful roadway illumination.

Realizing these capabilities in practice presents four key technical challenges. *First*, reliable streetlight detection in nighttime videos is complicated by small object sizes, varying viewing distances and angles, motion blur, and poor visibility. Visual confounders, such as vehicle headlights, traffic signals, illuminated signboards, and reflections, further complicate detection due to their resemblance to streetlights. *Second*, as the vehicle moves, the same streetlight appears across successive frames at varying scales and viewing angles. *Third*, classifying streetlights as *On*, *Off*, or *Dim* is challenging due to variations in ambient illumination, camera exposure, and interference from nearby light sources, particularly when distinguishing degraded output from normal operation. *Fourth*, quantifying useful roadway illumination requires estimating the useful illumination area of each luminaire, which depends on its characteristics, mounting geometry, and surrounding environmental conditions.

In this paper, we overcome many of these challenges and make the following specific contributions:

- We present the design and implementation of StreetLightIQ (SLIQ): an end-to-end edge-cloud streetlight monitoring system that integrates a lightweight detector (trained through knowledge distillation and optimized using INT8 quantization) with on-device temporal tracking on commodity smartphones, combined with cloud-based operational state classification and roadway illumination estimation.
- We develop a physics-based illumination model with counterfactual attribution that integrates streetlight geometry, operational state, lamp characteristics, road segmentation, and camera-motion estimates. By comparing modeled illumination with and without individual luminaires, the approach quantifies each lamp's served length, coverage and dark stretches under overlapping illumination and enables failure criticality ranking without calibrated lux measurements.
- We build an annotated nighttime driving dataset of 2,091 frames from six recordings covering highway, flyover, market and intersection scenes, with 2,528 lamp-head boxes, 25,279 confounding-light boxes, 12,458 ignore regions and 5,737 road and building polygons, together with a separately recorded 285-frame test recording.

2 Related Work

Existing automated streetlight monitoring approaches differ in their infrastructure requirements, spatial coverage, and granularity of their outputs. Centralized management systems enable remote fault reporting at the individual luminaire level, but require fixture-level instrumentation and networked controllers [1]. Satellite-based approaches support large-scale analysis of nighttime illumination but lack the spatial resolution needed to identify individual luminaires, determine their operational status, or quantify street-level roadway illumination [21]. In contrast, drive-by sensing enables finer grained observations without instrumenting individual luminaires, although its effectiveness depends on the sensing hardware and the reliability of observations collected from moving vehicles.

Within drive-by sensing, existing systems demonstrate complementary monitoring capabilities. Vehicle mounted platforms have combined illumination mapping with visual streetlight identification, height estimation, and geotagging [20], while lux-meter based approaches directly measure roadway illumination using specialized equipment [30]. Although these studies establish the feasibility of mobile streetlight monitoring, dedicated sensing hardware increases deployment costs and complexity. Smartphone based approaches offer a lower cost alternative. Prior studies have examined the feasibility and limitations of smartphone based nighttime measurements [16], while systems such as BikeL and NightLight demonstrate crowdsourced streetlight monitoring and ambient illumination mapping [4, 6]. However, ambient illumination measurements aggregate contributions from multiple light sources, making it difficult to attribute observed illumination to individual luminaires, particularly in visually cluttered urban environments.

Addressing this attribution challenge requires not only identifying individual luminaires but also reliably interpreting their visual observations. Recent vision-based approaches employ YOLO-based architectures with enhancements such as squeeze-and-excitation modules and image fusion to improve streetlight detection and fault recognition [3, 37]. Nevertheless, accurate detection alone does not establish each luminaire's contribution to roadway illumination. Estimating these contributions additionally requires temporal association and geometric interpretation of observations collected under varying camera motion, viewing angles, and exposure conditions. Nighttime environments further complicate these tasks through glare, competing light sources, occlusions, and motion blur [10, 29, 36]. SLIQ addresses this gap by integrating smartphone based visual sensing, temporal tracking, operational state classification, and physics-based illumination modeling to estimate individual lamp contributions to roadway coverage without requiring lux measurements at deployment.

3 System Overview

Figure 1 presents the SLIQ architecture, which integrates edge-based sensing and tracking with cloud-based visual analytics and illumination estimation. A dashboard mounted smartphone records frames along with gravity (accelerometer) and GPS data. Figure 5 shows a typical frame. The edge-cloud architecture distributes processing tasks according to their temporal and computational requirements. On the device, a distilled and quantized object detector identifies streetlight heads in video frames, while a custom tracker

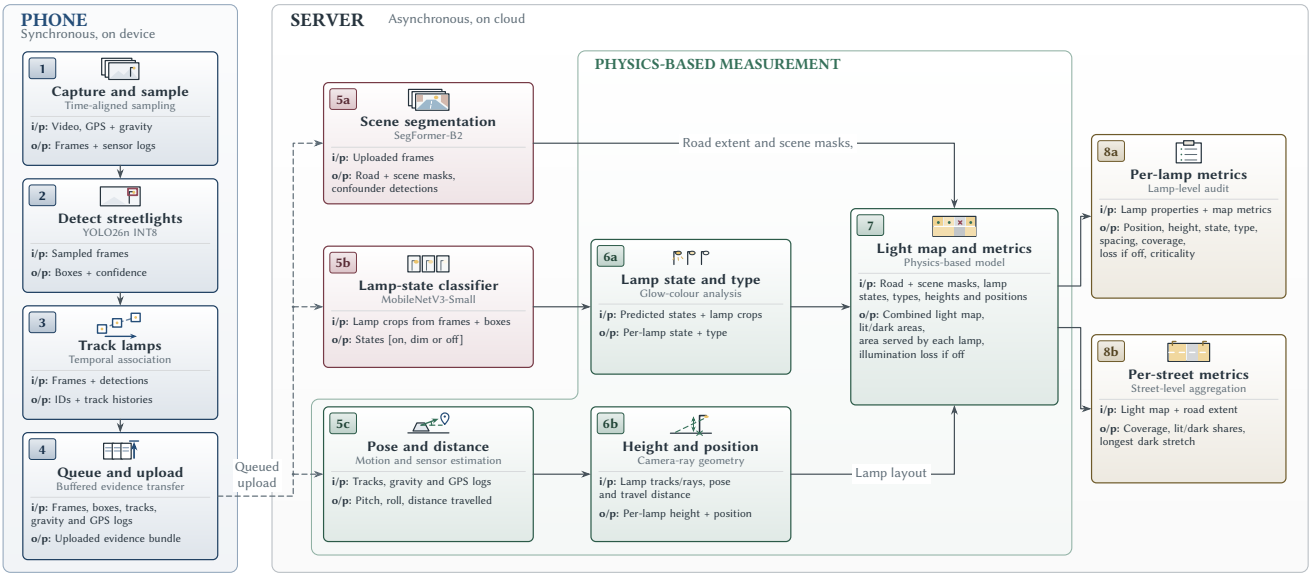


Figure 1: System architecture of SLIQ. ***Numbered markers show the processing order: video capture, synchronous on-phone detection and tracking, upload, asynchronous cloud processing, physical illumination modeling, and report generation.***

associates detections across successive frames to construct observation histories and reduce duplicate counting. These operations run concurrently with video capture: the camera records at its native frame rate, while the detector processes the most recent available frame, at most 5 per second, and skips the frames that arrive while it is busy. Performing them on-device avoids the storage and communication overhead associated with transmitting raw videos to the cloud.

The cloud asynchronously processes selected frames, lamp crops, and associated metadata for operational state classification and illumination estimation. A lamp state classifier estimates the probabilities of *On*, *Dim*, and *Off* for each observation, which are averaged across each track to determine the corresponding luminaire's operational state. In parallel, a semantic segmentation model classifies image pixels into four categories: background, road, building, and confounding light sources, such as vehicle headlights, illuminated signboards, and storefront lighting. The segmentation output is used to construct a usable road mask by excluding pixels affected by confounding illumination, saturation, regions below the dashboard line, and detected lamp head bounding boxes.

A physics-based model integrates tracked observations, operational state estimates, the usable road mask, lamp head glow color, and smartphone gravity (accelerometer) and GPS measurements to reconstruct streetlight geometry and estimate roadway illumination. Specifically, it estimates each luminaire's longitudinal and lateral position relative to the road, mounting height, and color category (white, orange or unclear), which sets an assumed beam shape. These estimates, together with the operational states, are used to construct a model-based illumination map of the roadway.

SLIQ reports metrics at two levels. At the *street level*, it reports the fractions of model-estimated lit and dark roadway and the longest continuous dark stretch. At the *lamp level*, it reports the road length served, illumination coverage, spacing to neighboring luminaires, and *loss-if-off*, the reduction in lit area if the lamp failed, computed by comparing the modeled illumination with and without it. Ranking lamps by loss-if-off gives a failure criticality ranking.

4 System Design and Implementation

4.1 On-Device Detection and Tracking

4.1.1 Lamp-Head Detection. We adopt knowledge distillation and quantization to obtain a lightweight detector for on-device deployment. Following an architectural comparison (Table 1), we select YOLO26m [18] (20.4 M parameters) as the teacher model and YOLO26n (2.4 M parameters) as the student. Both models predict tightly localized bounding boxes around streetlight heads.

Dataset and teacher training. Our dataset contains 1, 507/292/292 train/validation/test frames (1280 × 720) with 1, 894/397/237 lamp-head annotations. The data splits used approximately 30-second contiguous blocks separated by 112-frame gaps to limit temporal leakage. We annotate uncertain or non-localizable fixtures as ignore regions (8, 762/1, 085/791) and non-streetlight light sources as confounders (15, 489/3, 632/2, 693). An ignore-aware loss is implemented to mask classification penalties at unassigned prediction locations inside ignore regions while the confounders remain supervised as background. The COCO [23] pretrained YOLO26m teacher is fine-tuned for 60 epochs at 1, 280 × 720 with conservative augmentation (horizontal flipping and ±10% scaling). We disable color perturbations to preserve lamp appearance.

Student distillation. The COCO [23] pretrained YOLO26n student follows the same training schedule and supervision as the teacher. Each augmented image is first processed by the frozen teacher at 1, 280 × 720 and the student at 960 × 544. The teacher outputs are aligned to student grids. The objective combines the detection loss with three distillation terms: $mathcalL = \mathcal{L}_{\text{det}} + 4\mathcal{L}_{\text{score}} + 4\mathcal{L}_{\text{box}} +$

$\mathcal{L}_{\text{feat}}$. Score distillation uses disagreement-weighted binary cross-entropy masked in ignore regions. Box distillation uses GIoU on ground-truth-assigned locations with the teacher confidence above 0.3. Feature distillation aligns channel-standardized neck feature through learned 1×1 adapters which emphasize lamp regions, small object and stride-7 features.

Quantization and deployment. The checkpoints are selected based on validation map@50 followed by recall, precision and false-positives per frame. The operating threshold maximizes recall, subject to $\leq$ 0.2 false positives/frame. The student then undergoes static INT8 LiteRT quantization calibrated on 300 stratified training frames. The deployed 3.0MB detector is 93.2% smaller than the 44MB teacher model and achieves 377.5ms inference latency (2.65 fps). Tables 2 and 3 report the distillation and quantization comparisons.

4.1.2 Lamp Tracking. SLIQ associates lamp-head detections across frames with a lightweight custom tracker, producing one observation history per track for state aggregation (Section 4.2.1) and geometry estimation (Section 4.2.4). At each processed frame, the tracker must decide which of the N active tracks continues with which of the M new detections. Matching each track greedily to its nearest detection can let two tracks claim the same detection, so the tracker instead chooses the one-to-one pairing with the lowest total cost and solves this assignment problem exactly with the Hungarian algorithm [19].

Each track i stores its last box (center $\mathbf{c}_i$, width w_i, height h_i), an image-plane velocity $\mathbf{v}_i$ and the time Δt_i since it was last matched. All coordinates are normalized by the image width and height. The cost of pairing track i with detection j is

$$C_{ij} = 0.75\,\frac{d_{ij}}{g_i} + 0.10\,\Delta s_{ij}, \tag{1}$$

where $d_{ij} = \|\mathbf{c}_j - (\mathbf{c}_i + \mathbf{v}_i \Delta t_i)\|$ is the distance from the detection to the track's predicted position and $\Delta s_{ij} = |\ln(w_j/w_i)| + |\ln(h_j/h_i)|$ is the change in box size. The gate $g_i = 0.16 + \min(0.04\,\Delta t_i,\ 0.08)$ widens while a track goes unmatched, as its predicted position becomes less certain. Pairs with $d_{ij} > g_i$ or $\Delta s_{ij} > 4$ are infeasible. To let a track remain unmatched, the $N \times M$ cost matrix is extended with N dummy columns of cost 1, so a track keeps a detection only if $C_{ij} < 1$ and no cheaper overall assignment exists. Otherwise it is assigned a dummy column and counted as missed. After a match, the velocity is smoothed as $\mathbf{v}_i \leftarrow 0.6\,\mathbf{v}_i + 0.4\,(\mathbf{c}_j - \mathbf{c}_i)/\Delta t_i$, with each component clipped to ±1 per second to limit the effect of a wrong match.

General purpose trackers add IoU matching, Kalman filtering, appearance features, camera motion compensation and multi-stage confidence association [2, 5, 33, 38], whose computation is undesirable given the inference limited processing rate of the mobile detector. IoU matching is unreliable here because small lamp boxes can move more than their own width between frames. In SLIQ, the smoothed velocity predicts motion, the growing gate tolerates missed frames, and fragments that still occur are merged by position in the physics model (Section 4.2.4). Track identifiers therefore denote observation histories rather than unique luminaires. Each unmatched detection starts a new track, up to 48 concurrent tracks, without minimum hit confirmation, and tracks expire after 1.5 s without a match.

The tracker operates on every detector processed frame, with processing capped at 5 fps and reduced to 2 fps under severe thermal conditions. Track records are uploaded every 2 s. Upon track termination, the system uploads up to five full resolution frames selected across different apparent lamp sizes, together with the best lamp crop and one scene image, for subsequent cloud-based analysis.

4.2 Cloud-Based Analysis and Illumination Estimation

This section describes the cloud-based processing stages: operational-state classification, semantic segmentation, physics-based illumination modeling. The classifier processes lamp head crops extracted from uploaded frames using detector bounding boxes (Section 4.1.2), while the segmenter analyzes the full frames. The physics model integrates their outputs with lamp tracks and smartphone motion and location data to generate roadway illumination metrics.

4.2.1 Operational-State Classification. SLIQ estimates the operational state of each tracked luminaire from selected video frames using a cloud-based classifier. For each observation, a crop centered on the detected lamp head bounding box is expanded by a factor of 1.5 to capture the surrounding glow, pole, and contextual information. The crop is clipped to the image boundaries, resized to 224×224 pixels without preserving its aspect ratio, and normalized using the ImageNet mean and standard deviation. Brightness and contrast adjustments are deliberately excluded to preserve visual cues relevant to operational-state classification. The classifier employs MobileNetV3-Small [17], pretrained on ImageNet [8], with a three class linear classification head for *On*, *Dim*, and *Off* states.

Classifier training. Since naturally occurring *Dim* and *Off* luminaires are underrepresented in the dataset, the training set of 2,549 crops includes 250 synthetic *Dim* and 400 synthetic *Off* samples generated from real training images (Section 5.1). Validation is performed exclusively on real observations. The network is trained using cross-entropy loss with class weights proportional to the square root of inverse class frequency, yielding weights of 0.55, 1.29, 1.16 for *On*, *Dim*, *Off*, respectively. Class balanced sampling ensures equal representation across training batches, with real and synthetic samples contributing equally to *Dim* selections. Data augmentation includes horizontal flipping ($p = 0.5$), bounding-box shifts of up to 10% and rescaling by factors of 0.8–1.25 ($p = 0.3$ each), and Gaussian blur ($\sigma = 0.3$–0.8) and sensor noise of approximately two gray levels ($p = 0.1$ each).

Track-level state aggregation. Although the classifier produces a prediction for each observation, downstream illumination modeling requires a single operational-state estimate per luminaire. For lamp i, SLIQ aggregates the softmax probability vectors $\mathbf{p}_{i,t}$ across its observation history T_i, which is defined as:

$$\bar{\mathbf{p}}_i = \frac{1}{|T_i|} \sum_{t \in T_i} \mathbf{p}_{i,t} \tag{2}$$

The operational state is assigned to the class with the highest mean probability, $\hat{s}_i = \arg\max_{c \in \{\text{On,Dim,Off}\}} \bar{p}_{i,c}$. If the maximum mean probability is below 0.6, the luminaire is marked as uncertain. Aggregating predictions across observations reduces sensitivity to

individual frames affected by motion blur, transient occlusions, or other visual disturbances.

4.2.2 Scene Segmentation and Confounder Detection. SLIQ employs a semantic segmentation model to classify scene pixels into four categories: *background*, *road*, *building*, *confounding light*. The latter includes non-streetlight illumination sources, such as storefronts, illuminated signs, windows, and vehicle lights, which can interfere with roadway illumination estimation. The segmentation network uses SegFormer-B2 [35], initialized from a Cityscapes pretrained checkpoint [7], with its classification head replaced by a four class prediction head. The network produces pixel-wise logits at one-quarter of the input resolution, which are bilinearly upsampled to the original image dimensions.

Confounder detection and fusion. To improve the identification fo non-streetlight illumination sources, we train a separate COCO [23] pretrained YOLO26m detector. It uses the same 1, 507/292 training/validation frames, with 15, 489/3, 632 confounder annotations, including objects below 300 px^2. We treat eligible lamp heads as background with uncertain regions retaining the ignore-aware supervision of Section4.2.1. Training uses the same augmentation at 30 epochs at $1,280 \times 720$ resolution. However unlike the lamp detector, checkpoints are selected by validation confounding-light pixel IoU instead of bounding-box mAP, using masks generated by the process described below. The selected checkpoint (epoch 10) uses a confidence threshold of 0.03 which is tuned within the final segmentation fusion to prioritize recall, since missed confounders contaminate the usable-road.

Within each box, a bright core mask is obtained by selecting pixels whose grayscale intensity satisfies $g(x, y) \geq \max(180,\ 0.7 g_{\max})$, where $g_{\max}$ denotes the maximum grayscale intensity within the box. Morphological closing is subsequently applied within the bounding box boundaries. To accommodate small or diffuse light sources, the entire bounding box is retained when the bright core occupies at least 50% of the box or when the bright core, after a two pixel dilation used only for this size test, still contains fewer than 300 pixels. Independently of this test, the selected mask (bright core or whole box) is then dilated by two pixels to cover the halo around the light source. The detector and segmenter outputs are combined as follows. Each pixel is initially assigned the most probable class among *background*, *road*, *building*, after which pixels covered by the detector derived light mask are relabeled as *confounding light*. This ensures that detected non-streetlight illumination sources are explicitly excluded from subsequent roadway illumination estimation. The detector-derived mask overrides the network's confounding light prediction because the respective labels are themselves generated from annotated boxes by this rule. On validation, this raises confounding light IoU from 0.337 to 0.362 (Table 4), and taking the union of the two masks adds nothing further.

Segmentation training. The segmentation network is trained using randomly sampled 576×1024 pixel crops, with scaling factors between 0.75 and 1.15 and horizontal flipping with probability 0.5. To improve representation of underrepresented classes, half of the training crops are centered on pixels belonging to rare classes. The training objective combines class weighted cross-entropy and Dice loss [25]. The cross-entropy class weights are 0.71, 1.13, 2.06, 3.35 for *background*, *road*, *building*, *confounding light*, respectively. Pixels inside lamp-head boxes and below the dashboard line are ignore regions and are excluded from the loss. Pixel wise confidence weighting of cross-entropy loss, with lower weights near class boundaries and for box-derived light labels, yielded no validation improvement. The final model, therefore, uses uniform pixel weights.

Usable-road mask construction. The final output passed to the physics-based illumination model is a *usable road mask*, which identifies roadway pixels suitable for illumination estimation. A pixel is retained only if it is classified as *road*, lies more than five pixels from confounding light regions, is not saturated (all 8-bit color channels below 250), lies above the dashboard line, and falls outside detected lamp-head bounding boxes. By excluding pixels affected by unrelated light sources, saturation, and visual obstructions, the mask provides a filtered representation of the observable roadway for subsequent physics-based modeling.

4.2.3 Camera Geometry and Road Projection. It integrates lamp tracks, operational-state estimates, lamp head glow color, and smartphone gravity (accelerometer) and GPS measurements to reconstruct streetlight geometry, estimate roadway illumination, and derive illumination metrics. The usable road mask generated by the segmentation module defines the roadway region over which illumination is estimated.

The four segmentation classes serve distinct purposes in constructing this mask. *Road* is the only positive class, with its pixels projected onto the ground plane to estimate the lateral extent of the roadway. *Background* and *Building* serve as negative classes, preventing non-road regions from being included in the roadway illumination estimation, but they are not explicitly modeled as illuminated surfaces or occluders. The fourth class, *Confounding Light*, identifies non-streetlight light sources whose pixels are excluded to prevent interference with roadway illumination estimation. The mask is further refined by removing pixels within five pixels of confounding-light regions, saturated pixels, regions below the dashboard line, and detected lamp-head bounding boxes. The resulting filtered usable road mask provides the spatial domain for subsequent physics-based illumination modeling.

Camera geometry and coordinate system. SLIQ employs a local road aligned metric coordinate system with its origin at the camera, obtained by removing the camera's pitch and roll, where the x-axis points right, the y-axis points downward, and the z-axis aligns with the forward direction of travel. The road surface is approximated as a horizontal plane at $y = h_c$, where h_c denotes the camera height above the road. The camera follows a pinhole projection model with image dimensions $w \times h$ and horizontal field of view ϕ. Assuming equal horizontal and vertical focal lengths, the focal length in pixels is $f = \frac{w/2}{\tan(\phi/2)}$. The unnormalized camera ray corresponding to image pixel (u, v) is $\mathbf{r}_{\text{cam}}(u, v) = \left(\frac{u - w/2}{f}, \frac{v - h/2}{f}, 1\right)$. The camera height and field of view are set once for each smartphone and mounting configuration, not estimated from the video. In our deployment, $h_c = 1.3\,\text{m}$, $\phi = 70^\circ$, and the image resolution is 1280×720 pixels.

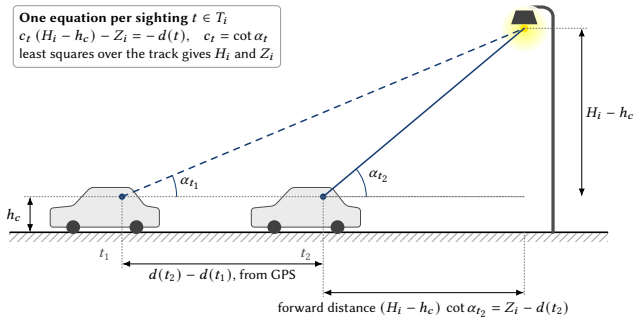


Figure 2: Lamp height from motion (in a side view that is not to scale). ***As the vehicle moves, the elevation angle α_t to a fixed lamp changes, and each sighting gives one linear equation in the height H_i and forward position Z_i of the lamp.***

Camera orientation and travel-distance estimation. Camera pitch and roll are estimated from smartphone gravity measurements. To suppress short-term fluctuations caused by vehicle vibration, braking, and turning, the gravity vector $\mathbf{g} = (g_x, g_y, g_z)$ is averaged over a window of 15 samples on either side of each video frame. The averaged vector is normalized and expressed in the device's landscape coordinate system, following the Android sensor convention (at rest the vector points away from the ground) with the phone in landscape left orientation and the rear camera looking along the device's $-z$ axis. The formulas below assume this convention. Pitch and roll are then computed as: $\theta_{\text{pitch}} = \arcsin\left(\text{clip}(g_z, -1, 1)\right), \theta_{\text{roll}} = \text{atan2}(-g_y, g_x)$, where atan2 is the four-quadrant arctangent. Each camera ray is transformed into the road aligned coordinate system using $\mathbf{r} = (r_x, r_y, r_z) = R_x(-\theta_{\text{pitch}})R_z(\theta_{\text{roll}})\mathbf{r}_{\text{cam}}$. For rays with $r_y > 0$, the intersection with the ground plane is obtained as $\mathbf{p} = \frac{h_c}{r_y}\mathbf{r}$. Only ground intersections within 80 m of the camera are retained to limit the effect of pitch errors, which grows with distance. The distance traveled along the vehicle trajectory is estimated from GPS measurements. Geographic coordinates are first converted into local metric coordinates using an equirectangular projection centered at the initial GPS fix. Speeds are computed between consecutive distinct fixes and smoothed using a rolling median over a ±3 s window to suppress spurious jumps of approximately 30–40 m caused by inaccurate GPS measurements. The resulting speeds are capped at 30 m/s and integrated over time to obtain the cumulative traveled distance: $d(t) = \int_0^t \tilde{v}(\tau)\, d\tau$, where $\tilde{v}(t)$ denotes the filtered vehicle speed.

4.2.4 Lamp Height and Position Estimation. SLIQ exploits the stationary nature of streetlights to estimate their mounting heights from observations collected as the vehicle moves (Figure 2). Let $\mathbf{r}_t$ denote the road aligned ray toward lamp i in frame t, and define $c_t = r_{z,t}/(-r_{y,t}) = \cot \alpha_t$, where α_t is its elevation angle (Figure 2). For a lamp mounted at height H_i, its forward distance from the camera is $(H_i - h_c)c_t$, where h_c denotes the camera height. Since the lamp is stationary, this distance also equals $Z_i - d(t)$, where Z_i is its initial longitudinal position and $d(t)$ is the cumulative distance traveled. Each observation therefore provides a linear constraint $c_t(H_i - h_c) - Z_i = -d(t)$ for $t \in T_i$, where T_i denotes the set of observations associated with lamp i. The resulting overdetermined system is solved using least squares to jointly estimate H_i and Z_i. To mitigate erroneous observations, two rounds of outlier rejection retain only sightings whose absolute residual does not exceed the greater of 1 m and 2.5 times the median absolute residual.

Height validation and fallback. Mounting height is fitted only for lamps with at least four valid observations, each at least four pixels from the image boundary and satisfying $-r_y > 0.02$, with a minimum distance traveled of 8 m between the first and last sightings. The fitted height must lie within 4-16 m. Otherwise, SLIQ assigns a representative street level height, which is computed as the median of fitted heights when at least three lamps have valid estimates, or a default of 9 m. Note that such lamps are explicitly marked as having an assumed height.

Lamp localization and track merging. Given the mounting height H_i, the three dimensional position of lamp i relative to the camera in frame t is obtained by intersecting its observation ray with the horizontal plane at the corresponding height $\mathbf{P}_{i,t} = \mathbf{r}_t \frac{H_i - h_c}{-r_{y,t}}$. The corresponding ground projected position in road coordinates is given as: $(s_{i,t}, x_{i,t}) = \left(d(t) + P_{z,i,t},\, P_{x,i,t}\right)$, where s and x denote longitudinal and lateral positions, respectively. This frame is straightened along the vehicle trajectory and does not compensate for turns, lane changes or curvature, which can bias lateral positions on bends. Since each lamp is placed from sightings about 30 m ahead, the assumption only needs to hold locally: on our drives, the GPS path deviated from a straight line by a median of 0.4 m over 30 m (90th percentile 2.4 m). Since angular errors produce larger localization errors at greater distances, the lamp position is estimated using the median of its three nearest observations, excluding those within 5 m, taken separately for the longitudinal and lateral coordinates. Lamps never observed within 30 m are flagged as distant observations. To account for fragmented tracks, lamp records whose estimated positions differ by at most 4 m in both longitudinal and lateral directions are merged into a single lamp record. Two tracks that were detected in the same frame as separate heads, with

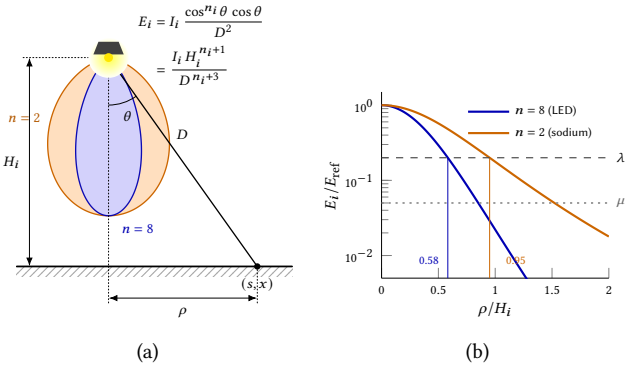


Figure 3: The light model. ***(a) Each lamp is a point source with a $\cos^n$ beam. Its horizontal illuminance at a road point depends on the distance D and the angle θ from the vertical. (b) Illuminance of a single On lamp at horizontal distance ρ from its foot, relative to E_{ref}, for a narrow LED beam ($n = 8$) and a broad sodium beam ($n = 2$), with $H_i = h_{\text{ref}}$. Vertical lines mark the radius within which the road is lit.***

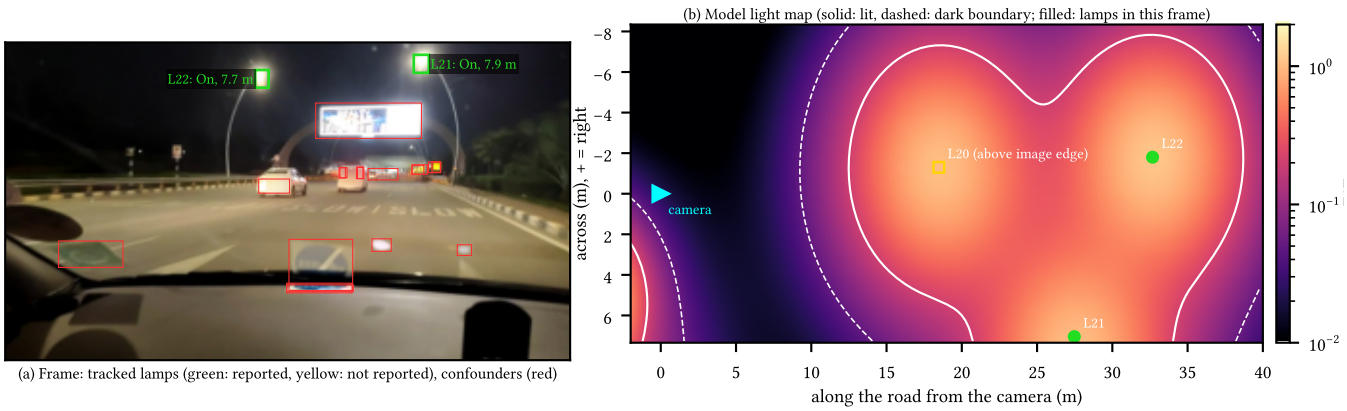


Figure 4: A real frame and the model light map of the same street. (a) A frame from a training drive, with the tracked lamps (fitted state and height) and detected confounders. (b) The light map of Section 4.2.6 from the camera to 40 m ahead. *Solid white lines bound lit road and dashed lines bound dark road. Lamp L20 lies above the image edge.*

centers at least one box width apart, are never merged, so closely spaced luminaires such as double-headed poles remain separate. Their positions and per-track mean operational-state probabilities are combined using sighting-count-weighted averages, and the lamp's state and intensity weight are computed from the merged probabilities, with fitted mounting heights taking precedence over assumed values.

4.2.5 Lamp Intensity and Beam Modeling. SLIQ assigns each luminaire a relative intensity weight based on the track-averaged operational-state probabilities obtained from Section 4.2.1. For lamp i, the intensity weight is defined as $I_i = \bar{p}_{i,\mathrm{On}} + \delta\, \bar{p}_{i,\mathrm{Dim}}$ with $\delta = 0.5$, where $\bar{p}_{i,\mathrm{On}}$ and $\bar{p}_{i,\mathrm{Dim}}$ denote the mean probabilities of the corresponding operational states. Thus, confidently classified *On*, *Dim*, *Off* lamps receive nominal intensity weights of 1, 0.5, 0, respectively. Because the weight uses mean probabilities rather than hard labels, a lamp with an uncertain state still contributes light in proportion to its probabilities. This is how uncertain luminaires enter the illumination map while being excluded from individual-lamp reporting (Section 4.2.6). These weights represent modeling assumptions rather than calibrated photometric measurements, as actual dimming characteristics vary across luminaires [31].

Lamp-type classification and beam modeling. SLIQ estimates the lamp type from the color of its surrounding glow, which informs the assumed beam distribution. For each detected lamp head, an annular region is defined between bounding boxes scaled by factors of 1.5 and 3. Within this region, only pixels whose maximum 8 bit sRGB channel value lies between 8 and 250 are retained, to exclude very dark and saturated observations, and the retained pixels are then converted from sRGB to linear RGB. At least 20 valid pixels are required to estimate the mean color. The color feature is defined as: $\kappa = \log\left(\frac{R}{B}\right)$, where R and B denote the mean linear red and blue intensities, respectively. A two class Otsu threshold [27], which picks the split that maximizes the variance between the two groups, estimated once from 380 training lamps, yields a decision boundary of $\kappa = 0.076$. Introducing a margin of 0.158 (a quarter of the distance between the two cluster means) on either side, lamps are classified as *White* (if $\kappa_i \le -0.082$), *Orange* (if $\kappa_i \ge 0.234$), and *Unclear* (otherwise). Because κ depends on the phone's white balance, the threshold may need to be re-estimated for a different camera. The inferred color category determines the beam shape exponent n_i used in the illumination model (the $\cos^{n_i}\theta$ term of Section 4.2.6, so a larger n_i gives a narrower beam): $n_i = 8$ for *White*, 2 for *Orange*, 4 for *Unclear* lamps. *White* lamps are modeled using a relatively narrow (LED like) beam, whereas *Orange* lamps are assigned a broader (sodium like) beam. Lamps with ambiguous color estimates use an intermediate beam exponent. These values are nominal modeling choices that approximate differences in beam spread rather than calibrated luminaire specific photometric characteristics.

4.2.6 Illumination Mapping and Audit Metrics. SLIQ models each luminaire as a point source located at road coordinates (s_i, x_i) and mounting height H_i, with its beam directed vertically downward. Following the inverse square and cosine laws, the relative horizontal illuminance contributed by lamp i at road position (s, x) is modeled as [9]:

$$E_i(s,x) = I_i \frac{\cos^{n_i}\theta \cos\theta}{D^2} = I_i \frac{H_i^{n_i+1}}{D^{n_i+3}} \tag{3}$$

$$D^2 = (s-s_i)^2 + (x-x_i)^2 + H_i^2, \qquad \cos\theta = \frac{H_i}{D} \tag{4}$$

Here, I_i denotes the relative intensity weight derived from the operational-state probabilities, while n_i controls the beam spread according to the inferred lamp type (Figure 3a). The aggregate illumination map is obtained by superposing contributions from all luminaires, and is given as: $E(s,x) = \sum_i E_i(s,x)$.

Road discretization. The illumination map is evaluated over a rectangular road strip whose lateral boundaries are estimated from usable road ground projections. Specifically, the strip width is determined by the 5th and 95th percentiles of the lateral coordinates of usable road points located 8-30 m ahead of the camera, aggregated across all frames. The lateral boundaries are restricted to ±12 m relative to the camera. If fewer than 2,000 valid points are available, a default width of 10 m, extending 5 m on either side, is used. Longitudinally, the strip begins at the earlier of 15 m before the

first lamp and the start of vehicle travel, and ends at the later of 15 m beyond the last lamp and 30 m beyond the end of travel. The resulting region is discretized into a grid of 0.5 m cells. This strip is modeled from road pixels (width) and from lamp positions and vehicle travel (length), not from the physical road boundary, and all coverage fractions are relative to it.

Relative illumination thresholds. SLIQ classifies roadway cells using thresholds defined relative to the modeled illumination directly beneath a fully operational lamp at the representative streetlight height h_{ref}, $E_{\text{ref}} = 1/h_{\text{ref}}^2$. A cell is classified as *lit* if $E(s,x) \geq \lambda E_{\text{ref}}$ where $\lambda = 0.2$, and *dark* if $E(s,x) < \mu E_{\text{ref}}$ where $\mu = 0.05$. Cells between these thresholds are classified as neither lit nor dark. Figure 3b illustrates the resulting lit radius for different beam widths. The lit-road classification serves as an operational proxy for useful roadway illumination [14, 22].

Street- and lamp-level metrics. SLIQ derives illumination metrics at two spatial scales. At the street level, it reports the fractions of lit and dark roadway cells, together with the longest continuous dark stretch along the road centerline. Because cells between the two thresholds are neither lit nor dark, the two fractions need not sum to one. At the lamp level, each luminaire i is assigned a served area A_i, comprising the roadway cells where its modeled illumination contribution exceeds that of every other lamp. The corresponding metrics are *served length* (longitudinal extent of the lamp's assigned roadway area), *coverage* (fraction of the served area classified as lit), *longest dark gap* (maximum continuous dark stretch along the road centerline within the lamp's served length), *spacing* (longitudinal distances to the preceding and succeeding luminaires on the same side of the road) and *loss-if-off* (reduction in illuminated roadway area if the lamp becomes non-operational). The loss-if-off metric quantifies the marginal contribution of each luminaire through a counterfactual analysis. Specifically, lamp i is removed from the illumination model while all other lamp contributions and the illumination threshold remain unchanged. The resulting loss is given as:

$$L_i = |\{(s,x) : E(s,x) \geq \lambda E_{\text{ref}}\} \cap \{(s,x) : E(s,x) - E_i(s,x) < \lambda E_{\text{ref}}\}| \quad (5)$$

where, $|\cdot|$ denotes the total area of the corresponding roadway cells. The metric is reported in square meters and as a fraction of A_i. Ranking luminaires by L_i provides a *failure criticality ranking*, prioritizing lamps whose outages would cause the greatest reduction in illuminated roadway area. The ranking describes the consequence of a failure rather than its likelihood: it does not predict which lamps are likely to fail. Lamps already classified as *Off* contribute negligible modeled illumination and consequently have near-zero loss-if-off values. Their absence is instead reflected in the existing dark regions. Figure 4 illustrates the per-lamp metrics, while Table 6 summarizes the aggregate results from the recorded drives.

Reporting criteria. Individual-lamp metrics are reported only when four reliability criteria are satisfied: (i) the mounting height is directly fitted rather than assumed, (ii) the operational state is not uncertain, (iii) the lamp has been observed within 30 m, and (iv) at least three sightings are available. Of the 396 localized luminaires in the recorded drives, 80 satisfy these criteria (Table 6). All localized lamps contribute to the street-level illumination map, regardless of whether their individual metrics satisfy the reporting criteria. For each street, SLIQ outputs the light map, the street metrics and an inventory of the luminaires with their position, height, state, type and lamp-level metrics.

5 Evaluation

We first describe our dataset (Section 5.1) and then evaluate SLIQ in four stages: detection, classification and segmentation, perception and on-device deployment, and the physics model.

5.1 Dataset Construction

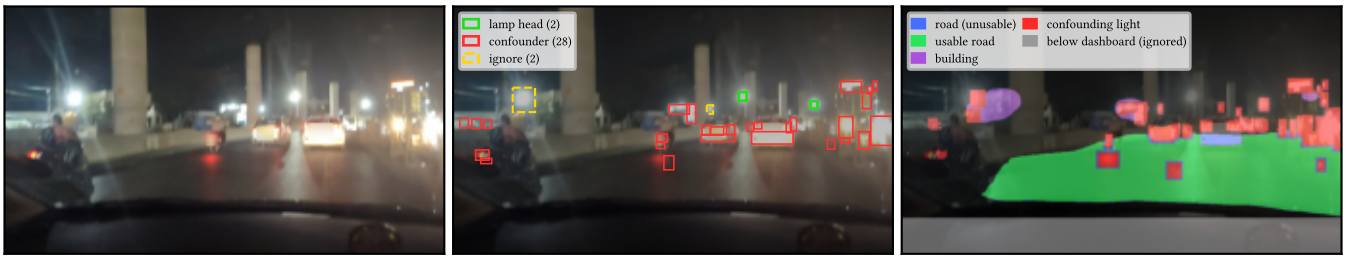
The principal dataset was collected across six nighttime recording runs using a car mounted smartphone camera. The recordings were divided into contiguous 30 s clips separated by 2 s gaps to reduce temporal leakage. Each clip was assigned a unique identifier and restricted to a single data partition. The independent recording described below provides a test that shares no clips with the principal dataset. Frames were extracted at 3 fps (yielding 2, 091 frames distributed into 1, 507 training, 292 validation, 292 held-out test frames).

Separately, an independent seven minute nighttime recording was collected for evaluation of the frozen perception pipeline. This recording yielded 285 sampled frames containing 298 annotated lamp observations and was not used for training or model selection. We distinguish this independent evaluation footage from the 292-frame test partition of the principal dataset. Unless otherwise specified, results described as obtained on separately recorded test footage refer to the independent 285-frame recording.

The physics model is evaluated on video spans from the six recordings of the principal dataset. Each span covers the annotated frames of one data split, with a lead-in of up to 10 s. Frames are sampled at 5 fps, together with every annotated frame. The training split has 16 spans (705 s), the validation split 6 spans (135 s) and the test split 6 spans (170 s). We refer to the spans of the test split as the held-out test spans.

To test the light model against measured illuminance, we use published road-surface measurements from a street-lighting evaluation in Groton, Connecticut [26]. The report gives 268 readings for four installations, taken at 6.1 m intervals along the road on several lines across it. Our transcription reproduces the street averages published in the report.

Each frame contains annotations for the detector, classifier and segmentation. Detector boxes were tightly drawn around visible lamp heads. Bounding boxes excluded supporting poles, mounting structures, and surrounding glow. Identifiable fixtures falling below a minimum area threshold of 300 sq. px were also strictly excluded. The classifier uses lamp crops, counted separately from the detector boxes as they include some lamps that are ignore regions for the detector. The principal dataset contains 2,528 detector-eligible lamp-head boxes, distributed as 1,894 in training, 397 in validation and 237 in test. These counts refer specifically to the boxes used for detector evaluation and exclude the separately recorded evaluation footage and the annotated ignore regions. Total crops enclosing an 'On' streetlight were 1824 in train, 406 in validation and 226 in test. Total crops enclosing a 'Dim' lamp were 75 in train, 24 in validation and 8 in test. Only 2 crops with 'Off' lamps were available which were assigned to test. In total, the principal dataset contains 2,528



(a) Raw frame (b) Annotations (c) Predicted segmentation and usable road mask

Figure 5: A typical nighttime frame from the validation split. (a) Raw frame. (b) Annotations: lamp heads, confounding light sources and ignore regions. (c) Output of the deployed segmenter, with the usable road mask passed to the physics model.

lamp-head boxes, 8,087 small-lamp and 4,371 other ignore regions, 25,279 confounding-light boxes and 5,737 scene polygons (Figure 5).

To smooth the class imbalance, we carefully synthesized a total of 250 artificial 'Dim' lamps and 400 artificial 'Off' lamps. The synthetic samples were generated from actual data to preserve visual characteristics of original recordings that include lamp geometry, background context, camera noise and exposure conditions. This procedure modified the apparent emission intensity of the lamp head and its associated illumination in the surrounding environment. We used an exponential spatial decay function to model the attenuation of illumination with increasing distance from the lamp. This produced a gradual transition between illuminated and non-illuminated regions. For 'Dim' samples, the lamp retained weak visible emission with a corresponding reduced illumination of its surroundings, and for 'Off' samples, direct emission and its associated illumination were suppressed. All synthetic samples were restricted to the training partition. Validation and test performance was evaluated exclusively on real observations. For the classifier in total, there were a total of 2549 crops in train, 430 in validation and 236 in test.

For segmentation, each pixel was assigned to one of four classes: background, road, buildings and confounding lights. For training, the total share of labeled pixels was: 63.9% were background, 25.5% were road, 7.7% were buildings and 2.9% were confounding lights. A similar split was observed in the validation and test splits. Pixels that covered lamp-head boxes and everything below the car-dashboard line were ignored. These ignored pixels comprised nearly 21% of all pixels. The annotated dataset is available on request for research use. Faces and vehicle number plates will be blurred before release.

We first report the performance of our base teacher model (YOLO26m), compare it with YOLO11, YOLO12 and RF-DETR, and justify the teacher selection with results from three random seeds. We then report the distilled and quantized variants, followed by the classifier and segmenter on validation and test data.

5.2 Detection

We compared four medium-sized detectors (YOLO26m, YOLO11m, YOLO12m and RF-DETR Medium) under an identical training recipe (Table 1), and selected YOLO26m as our teacher. Only this model was subsequently distilled and quantized. Over three seeds with the final training setup, YOLO26m reached a validation mAP50 of 0.858 ± 0.039, and we kept the best seed as the teacher. On validation, it achieved a **mAP50** of **0.895**, **mAP50–95** of **0.624**, **recall** of **0.821**, **precision** of **0.869** and **0.17 false positives per**

Table 1: Detector selection on validation under the model-selection setup (1280 px unless stated). P and R at each run's validation-chosen confidence threshold.

Model	Params	mAP50	mAP50–95	P	R	Train (h)
YOLO26m (3 seeds)	20.4M	0.858±0.039	0.541±0.072	0.852	0.750	1.8
best seed		0.895	0.624	0.869	0.821	1.9
YOLO11m	20.1M	0.848	0.547	0.850	0.761	1.5
YOLO12m	20.1M	0.856	0.549	0.867	0.778	3.2
RF-DETR (Medium)	–	0.771	0.520	0.813	0.681	2.1
YOLO26m @ 640 px	20.4M	0.750	0.495	0.824	0.713	0.8
YOLO26m @ 1600 px	20.4M	0.858	0.493	0.832	0.715	1.5

Table 2: Knowledge distillation on validation (PyTorch FP32).

Model	Training	Input	mAP50	mAP50–95	P	R	FP/f
YOLO26m	teacher	1280	0.858±0.039	0.541±0.072	0.852	0.750	0.17
YOLO26s	plain	960	0.859±0.007	0.485±0.015	0.849	0.770	0.19
YOLO26s	KD	960	0.881±0.017	0.559±0.022	0.874	0.810	0.16
YOLO26s	plain	1280	0.869	0.529	0.848	0.773	0.19
YOLO26s	KD	1280	0.907	0.628	0.854	0.851	0.20
YOLO26n	plain	960	0.800±0.035	0.450±0.029	0.843	0.675	0.17
YOLO26n	KD	960	0.868±0.026	0.523±0.029	0.855	0.804	0.19
YOLO26n	plain	1280	0.840	0.544	0.862	0.786	0.17
YOLO26n	KD	1280	0.869	0.582	0.847	0.783	0.19

Table 3: Quantization of the distilled students (544×960 input) on validation and on the separately recorded test footage, at each file's validation-chosen confidence threshold. Size: exported LiteRT file.

		Validation			Test footage			
Format	Size (MB)	mAP50	R	FP/f	mAP50	R	FP/f	CPU (ms)
YOLO26m teacher								
PyTorch FP32	–	0.895	0.821	0.17	0.806	0.809	0.15	–
YOLO26n KD student								
PyTorch FP32	–	0.897	0.849	0.17	0.801	0.805	0.18	37
LiteRT FP32	9.9	0.898	0.849	0.17	0.801	0.805	0.18	35
LiteRT FP16	5.0	0.898	0.849	0.17	0.801	0.805	0.18	47
INT8 dynamic-range	2.9	0.889	0.836	0.20	0.783	0.799	0.18	57
INT8 static (deployed)	3.0	0.869	0.786	0.16	0.778	0.792	0.19	60
YOLO26s KD student								
LiteRT FP32	38.2	0.900	0.841	0.17	–	–	–	89
INT8 static	10.3	0.884	0.816	0.14	–	–	–	117

frame at its validation-chosen confidence threshold of 0.15, with a median **center error** of **0.025**. We define center error as the distance between the centers of a matched predicted box and its ground-truth box, divided by the diagonal of the ground-truth box (0.74 px in absolute terms). On separately recorded footage (285 frames, 298 lamps), scored once after the model was frozen, it reached 0.806 mAP50, 0.849 precision, 0.809 recall and 0.15 false positives per frame: recall and the false-alarm rate carry over from validation almost unchanged, and the drop in mAP50 comes mostly from box precision (mAP50–95 0.417) rather than missed lamps. Of the 57 lamps missed on this footage, 25 are motion-blur streaks, 19 are wall-mounted fixtures or lamps inside underpasses, 10 are found but with a box just below 0.5 IoU, and 3 are other cases. Of the 43 false positives, 16 are real lamps that annotators left unlabeled, 17 are box-placement disagreements, 9 are duplicate boxes on one lamp, and only one is a genuine confounding light source.

To run on the phone, we distilled the teacher into YOLO26s and YOLO26n students using output distillation on class scores and boxes together with feature distillation on the neck (Table 2). Distillation improved both students over identical training without a teacher: at 960 px, YOLO26n rose from 0.800±0.035 to 0.868±0.026 mAP50 over three seeds, mostly through recall (0.675 to 0.804), and YOLO26s rose from 0.859±0.007 to 0.881±0.017. The best YOLO26n student matches the teacher on validation (0.897 vs. 0.895 mAP50) with about a ninth of its parameters, so we deploy it. We then exported it to LiteRT at a 544×960 input (Table 3). FP16 halves the model to 5.0 MB with no loss. Static INT8, with the detection head's concatenation operators kept in float, shrinks it to 3.0 MB at a cost of 0.028 validation mAP50. On the same separately recorded footage, the deployed INT8 model reaches 0.778 mAP50, 0.811 precision and 0.792 recall at 0.19 false positives per frame, close to the FP32 teacher's 0.809 recall and 0.15 false positives per frame.

5.3 Classification and Segmentation

We use a fine-tuned MobileNetV3-Small as our lamp-state classifier (Section 4.2.1). It is trained with AdamW [24] at a learning rate of 10^{-4}, weight decay of 10^{-4} and batches of 64. The learning rate warms up over five epochs, and only the head is trained for the first five epochs. Training runs for at most 50 epochs and stops early after 10 epochs without improvement. We keep the checkpoint with the lowest class-weighted cross-entropy on the validation crops, breaking ties by the mean of On and Dim recall. Synthetic Dim and Off crops are used only for training, and validation and test use real crops only. On validation (430 crops), the classifier labels 0.907 of lamps correctly. On the held-out footage, accuracy is 0.856 (255 of 298 lamps).

The segmenter of Section 4.2.2 combines a SegFormer-B2 network with a separate YOLO26m confounder detector, whose detector-derived evidence is used to refine the confounding-light segmentation during inference. The network is trained for 30 epochs. The checkpoint is chosen on validation by confounding-light IoU, subject to minimum road and building IoU, and the detector's confidence threshold (0.03) is also chosen on validation. All design choices were made on validation, and the test footage was scored once, after the model was frozen.

Table 4: Segmentation ablation on validation, and the final model on the test footage.

Configuration	mIoU	Road	Building	Conf. light
Model and training (validation)				
MobileNetV3 + DeepLabv3 (3 seeds)	0.523±0.003	–	–	0.333±0.010
SegFormer-B2	0.536	0.692	0.345	0.334
SegFormer-B5 backbone	0.527	0.698	0.323	0.321
full-frame crops (720×1280)	0.538	0.695	0.360	0.333
streetlights as background	0.532	0.688	0.337	0.336
ignore sparse-building packs	0.496	0.639	0.313	0.311
Inference (validation)				
SegFormer-B2 (3 seeds)	0.529±0.015	0.682	0.356	0.329
deployed seed	0.542	0.692	0.377	0.335
+ flip averaging	0.543	0.692	0.376	0.337
+ confounder detector (final)	**0.547**	0.693	0.369	**0.362**
Test footage (285 frames)				
Final segmenter	**0.607**	0.690	**0.450**	**0.492**

Table 5: On-device detector performance on a Samsung Galaxy A16 5G (LiteRT, CPU, four threads, 960×544 input).

Detector	Latency (ms) ↓	Pipeline FPS ↑	Memory (MB) ↓
YOLO26n FP32	598.0	0.9	90
YOLO26n INT8 dynamic-range	503.6	1.2	82
YOLO26n INT8 static (deployed)	**377.5**	**1.3–1.5**	**72**
YOLO26s INT8 static	639.2	0.9	105

Table 4 ablates these choices on validation. Replacing the MobileNetV3 + DeepLabv3 baseline with SegFormer-B2 raises mIoU from 0.523 ± 0.003 to 0.536, mainly through buildings. A larger backbone (SegFormer-B5), full-frame training crops and training streetlights explicitly as background change mIoU by less than the seed spread of SegFormer-B2 (±0.015), so we keep the simpler configuration. Ignoring background in training packs where buildings are sparsely labeled lowers mIoU to 0.496, so we keep all labels. Across three seeds, SegFormer-B2 reaches 0.529 ± 0.015 mIoU, and the seed we deploy reaches 0.542. The confounder detector gives the largest single gain in confounding-light IoU (0.337 to 0.362), for a final validation mIoU of 0.547.

On the test footage (285 frames), the final segmenter reaches 0.607 mIoU, compared with 0.581 for the baseline. Building IoU rises from 0.385 to 0.450 and confounding-light IoU from 0.464 to 0.492, with confounding-light recall rising from 0.656 to 0.806, while road IoU is unchanged at 0.690. The test footage scores higher than validation mainly through the building and confounding-light classes, while road IoU is the same on both. Confounding-light labels are derived from the annotated boxes by the rule of Section 4.2.2, so confounding-light IoU measures agreement with that rule. The bright-road share, defined as the fraction of road pixels with luma ≥ 80, computed from predicted masks tracks the share computed from ground-truth masks with a correlation of 0.96 and a mean absolute difference of 0.066.

5.4 Perception and Deployment Evaluation

We evaluate the detection, lamp-state classification and segmentation components on independently collected footage comprising 285 frames and 298 annotated lamp observations. This recording was not used for training or model selection, and the models were

Table 6: Summary of the road and lamp metrics from the physics-model applied to the recorded drives.

Quantity	Value
Road covered	8.2 km
Lamps placed	396 (372 On, 24 Dim, 0 Off)
Lamp type	White: 220, orange: 44, unclear: 132
Lamps with fitted height	165 (median height: 9 m)
Lamps with all metrics	80
Road strip width	13.9 m
Median spacing to next lamp on same side	24 m
Median served length	26 m
Median coverage	0.29
Median loss-if-off share	0.21

evaluated after their configurations had been frozen. The teacher detects 241 of the 298 lamps (recall 0.809, 0.15 false positives per frame), and the deployed INT8 student reaches a recall of 0.792 (Table 3). Given ground-truth lamp boxes, the classifier assigns the correct state to 255 of the 298 lamps (0.856). With the teacher's boxes, 214 of the 298 lamps are both detected and correctly classified (0.718).

We deployed the app on a Samsung Galaxy A16 5G (SM-A166P). The detector runs on the CPU (four threads, XNNPACK) at a 960×544 input, and Table 5 reports its latency from survey sessions with each variant. Alongside video, the app logs GPS fixes and gravity (accelerometer) readings, which the physics model uses for distance traveled and camera tilt. Static INT8 is the fastest variant at 378 ms per frame, with 37% lower latency than FP32. The detector alone would allow about 2.65 frames per second. The end-to-end rate of 1.3–1.5 frames per second also includes the other on-device work for each frame. The detector's runtime memory, measured with the same model files and runtime on a workstation, is 72 MB for the deployed model. The battery drained at 14.7% per hour, so a full charge lasts about 6.8 h, and battery temperature stayed between 31.6 and 34.5 °C. In our recordings the vehicle averaged about 35 km/h and passed about 45 lamps per kilometer, so a single phone could survey roughly 240 km of road on one battery charge. No inference failures were recorded.

5.5 Physics Model Evaluation

We evaluate the physics model in three complementary ways. Tests with known lamp layouts on the real drives measure how accurately the geometry and failure-criticality rankings are recovered. A repeatability test on real lamps checks how consistently heights are recovered from real observations. Published road-surface measurements test the light model itself. All choices were made on the training and validation spans, and the held-out test spans were scored once.

5.5.1 Lamp Geometry. We first test the height fit against known answers. For every real track in the training and validation spans, the lamp is replaced by a synthetic one at a known height and position, keeping the real camera poses, distances and frames. Its box is projected with 2 pixels of noise and 0.5° of per-frame tilt noise, and the height fit of Section 4.2.4 then runs unchanged. Over 261 lamps, the median height error is 0.23 m, 97% of heights are within 1 m and all are within 2 m. The median along-road position error is 0.45 m. Without noise the recovery is exact.

We also check repeatability on 137 real lamp observation histories from the training and validation drives, each with at least 8 sightings over at least 16 m of travel, fitting the height separately from the even and the odd sightings. The two halves agree to 0.76 meters in the median with a correlation of 0.85. If the errors of the two halves are independent and normally distributed, a fit to all sightings has a median error of about half this disagreement, 0.38 m. This is close to the 0.35 m height error of the end-to-end test in Section 5.5.2.

Table 7: End-to-end estimation performance against reference-model outputs for known lamp layouts on 80 road segments, with lamps sighted at the tracker's 5 fps processing cap, at the detector's standalone rate (2.65 fps) and at the end-to-end rate measured on our test phone (1.5 fps)

Scope	Metric	5 fps	2.65 fps	1.5 fps
Road-level				
Failure criticality	Spearman correlation / top-three critical-lamp hit rate	0.94 / 0.98	0.93 / 0.98	0.90 / 1.00
Dark stretches	Accuracy / precision / recall	0.87 / 0.89 / 0.93	0.85 / 0.88 / 0.92	0.81 / 0.86 / 0.89
Model-lit road	Accuracy	0.92	0.90	0.88
Model-lit street share	Spearman correlation / median error	0.93 / 0.013	0.94 / 0.013	0.91 / 0.014
Longest dark stretch	Spearman correlation / median error	0.77 / 8.8 m	0.75 / 9.3 m	0.71 / 12.8 m
Per-lamp				
Lamp height	Median error	0.35 m	0.36 m	0.44 m
Lamp position	Median error	1.25 m	1.40 m	1.78 m
Lamp state	Accuracy	0.90	0.90	0.89
Lamps found	Precision / recall	0.83 / 0.76	0.77 / 0.70	0.65 / 0.58

5.5.2 End-to-End Evaluation Against Known Layouts. We evaluate the pipeline using a known-answer protocol: lamp layouts with known positions, heights and states are embedded into the held-out drives, using their trajectories, frames and GPS. Poles stand every 25–45 m on one or both sides of the road, 4–9 m from the camera line and 7–12 m high, with white LED heads. Their states are On (80%), Dim (12%) and Off (8%). The illumination metrics are computed using the same light model from the true layout, so this test measures how well the pipeline recovers that reference under realistic perception errors. The classifier behavior is also real. The pipeline then processes these lamps as the deployed system would. Because an unlit lamp head is rarely visible at night, the test assumes the hardest case, in which Off lamps are never detected. Their failure must then be inferred from the gap they leave in the light map, as in the deployed system. A lamp is seen only while it is 8–80 m ahead and inside the frame, 25% of sightings are missed, 10% of tracks are broken, and boxes carry 2 px of noise. The class probabilities of each lamp come from a real lamp with the same true state. Table 7 reports the results.

On the test spans, the failure-criticality ranking produced by the pipeline has a median Spearman correlation of 0.94 with the reference ranking, and the lamp it ranks most critical is among the three most critical reference lamps in 98% of the road segments. Dark stretches on the center line are found with an accuracy of 0.87 and lit road stretches are found with an accuracy of 0.92. The lit share of a street also follows the true one closely with Spearman of 0.93 and median error of 0.013 but the longest dark stretch is harder to recover with Spearman of 0.77 and median error of 8.8 meters. Lamp heights have a median error of 0.35 m and positions 1.25 m. The model also finds 76% of working lamps and assigns

Table 8: Sensitivity of per-lamp coverage and loss-if-off to changes in model assumptions.

Changed assumption	Coverage		Loss-if-off	
	Rank correlation	Change	Rank correlation	Change
LED exponent 8 → 4	0.91	0.01	0.85	0.02
LED exponent 8 → 16	0.95	0.00	0.90	0.01
All lamps broad beam ($n = 2$)	0.82	0.12	0.67	0.14
Lit threshold $\lambda : 0.2 \rightarrow 0.1$	0.98	0.08	0.94	0.07
Lit threshold $\lambda : 0.2 \rightarrow 0.4$	0.95	0.09	0.79	0.07

Table 9: Prediction error on held-out measurement lines across the road.

Installation	Model	Interpolation
Street 1, high-pressure sodium	**32%**	46%
Street 1, induction	**19%**	37%
Street 2, high-pressure sodium	**22%**	36%
Road-edge lines (5)	**34%**	70%
Interior lines (8)	**22%**	31%
All 13 held-out lines	**23%**	37%

the correct state to 90% of them. These results are close to those on the training and validation spans (failure-criticality Spearman correlation 0.93, top-three hit rate 95%, height error 0.34 m). The on-device detector alone processes 2.65 frames per second (1/377.5 ms), so we also evaluate the pipeline with lamps sighted only at that rate. The estimation results remain close to the 5 fps result: the failure-criticality ranking has a Spearman correlation of 0.93, the lit share of a street 0.94 and lamp heights a median error of 0.36 m, while the share of working lamps found falls from 76% to 70%. At 1.5 fps the rankings and street-level shares still hold (Spearman correlations of 0.90 and 0.91), but fewer lamps collect enough sightings for a height fit, so the model finds 58% of working lamps and the median error of the longest dark stretch rises to 12.8 m. Because the deployment model anticipates repeated passes over the same streets, a lamp missed on one pass can be observed on later ones.

5.5.3 Sensitivity to Change in Assumptions. The model relies on fixed constants for beam exponents, lit threshold and strength of dim lamps. To keep the test spans untouched, we re-run the pipeline on the training and validation spans with each assumption changed, and compare the per-lamp results with the frozen configuration over the 75 lamps whose metrics are reported. We find that the ranking of lamps is stable under reasonable change.

Narrowing or widening the LED beam and halving or doubling the lit threshold keeps Spearman correlations between 0.79 and 0.98 for both coverage and loss-if-off. Changing Dim strength from 0.5 to 0.3 or 0.7 has no effect, therefore it is not mentioned in Table 8. We do notice that change in beam type causes a difference as treating every lamp as broad beam lowers rank correlation of loss-if-off to 0.67 which is why each lamp's type is estimated from its glow color. Lamps failing the reporting criteria still enter the street map with an assumed height and a probability-weighted intensity. Redrawing each lamp's state from its classifier probabilities and each height from the fitted-height distribution (fitted heights jittered by repeatability-test noise), 200 times per training and validation street, changed the lit share by a median of 0.003 (90th percentile 0.012, below the pipeline's lit-share error in Table 7) and the dark share by 0.007 (0.028), and preserved the street ranking (median Spearman 0.99). The longest dark stretch varied more (1 m, 90th percentile 58 m), as one changed height can join or split dark stretches.

5.5.4 Comparison with Measured Illuminance. We test the light model of Section 4.2.6 against these measurements. Lamp positions come from the published pole stations. The report provides nominal installation mounting heights, while lamp-specific heights and the remaining unobserved model parameters are fitted using the other measurement lines. The held-out line across the road is then predicted. This tests the form of the light model once its parameters are fitted, separately from the geometry that the SLIQ estimates from video. As a baseline, each held-out point is interpolated as the mean of the other lines at the same position along the road, so the baseline uses real measurements at that position. Both use only the other lines of the same installation. Error is the mean absolute difference between predicted and measured illuminance on a held-out line, divided by that line's mean measured illuminance, with the median taken over lines. We report the three high-pressure sodium and induction installations. Over their 13 held-out lines, the model's median error relative to mean illuminance is 23%, against 37% for interpolation, and 68% of its predictions fall within a factor of 1.5 of the measurement, against 51% (Table 9). The gain is largest at the road edges, where interpolation can only copy the middle of the road. There the model's error is 34%, against 70%.

6 Limitations

SLIQ's estimated illumination values are relative and not calibrated lux measurements. The light model treats each lamp as a downward-facing point source with a symmetric $\cos^n$ beam, so it cannot represent asymmetric optics or occlusions, limiting the reliability of estimated coverage. Modeling these would require per-luminaire photometric profiles and orientations, or lux measurements, which drive-by video cannot provide without additional instrumentation. Our evaluation suffers from scarcity in *Dim* and *Off* observations. Lamps with very weak or no emission are also intrinsically difficult to detect, so Off lamps are inferred only from the gaps they leave in the light map. Municipal inventories could resolve such ambiguities, but they require supplementary data collection beyond nighttime drive-by sensing, which is beyond the scope of this work.

Detection completeness is limited by the measured on-device processing rate of approximately 1.5 fps (58% of working lamps recovered). Coverage could be improved by repeated traversals, which are expected in the proposed deployment but have not been measured explicitly.

7 Conclusion

We presented SLIQ, a drive-by streetlight monitoring system that integrated distilled, INT8-quantized on-device detection and temporal tracking with cloud-based operational state classification and scene segmentation. We developed a physics-based model that estimated the relative roadway illumination and quantified contributions of each individual lamp through a counterfactual analysis, thus enabling failure-criticality rankings without requiring calibrated lux

measurements. We constructed an annotated nighttime driving dataset and evaluated our system's performance.

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A Tracker Algorithm

Algorithm 1 gives one update of the on-device tracker of Section 4.1.2. It runs once per detector-processed frame. Coordinates are normalized by the image width and height.

Algorithm 1 Per-frame update of the SLIQ tracker

Require: active tracks $\mathcal{T}$, each with box center $\mathbf{c}_i$, size (w_i, h_i), velocity $\mathbf{v}_i$ and last match time τ_i; detections $\mathcal{D} = \{(\mathbf{c}_j, w_j, h_j)\}_{j=1}^{M}$ at time t

for each track $i \in \mathcal{T}$ with $t - \tau_i > 1.5$ s **do**
  remove i from $\mathcal{T}$ and finalize it (upload frames)
end for
$N \leftarrow |\mathcal{T}|$; build an $N \times (M+N)$ cost matrix C
for each track i and detection j **do**
  $\Delta t_i \leftarrow t - \tau_i$; $g_i \leftarrow 0.16 + \min(0.04\,\Delta t_i,\ 0.08)$
  $d_{ij} \leftarrow \|\mathbf{c}_j - (\mathbf{c}_i + \mathbf{v}_i \Delta t_i)\|$; $\Delta s_{ij} \leftarrow |\ln(w_j/w_i)| + |\ln(h_j/h_i)|$
  if $d_{ij} > g_i$ **or** $\Delta s_{ij} > 4$ **then**
    $C_{ij} \leftarrow \infty$ ▷ infeasible pair
  else
    $C_{ij} \leftarrow 0.75\, d_{ij}/g_i + 0.10\, \Delta s_{ij}$
  end if
end for
$C_{i,M+k} \leftarrow 1$ for all i and $k = 1, \ldots, N$ ▷ "no match"
$a \leftarrow \textsc{Hungarian}(C)$ ▷ minimum-cost one-to-one assignment
for each track i **do**
  $j \leftarrow a(i)$
  if $j \le M$ **and** $C_{ij} < 1$ **then**
    $\mathbf{v}_i \leftarrow \mathrm{clip}\big(0.6\,\mathbf{v}_i + 0.4\,(\mathbf{c}_j - \mathbf{c}_i)/\Delta t_i,\ -1,\ 1\big)$
    $(\mathbf{c}_i, w_i, h_i) \leftarrow (\mathbf{c}_j, w_j, h_j)$; $\tau_i \leftarrow t$
    append detection j to the history of i
  else
    track i is missed in this frame
  end if
end for
for each detection j not assigned to a track **do**
  if $|\mathcal{T}| < 48$ **then**
    start a new track from j with $\mathbf{v} = \mathbf{0}$, $\tau = t$
  end if
end for

B Light-Map Metrics on an Illustrative Street

Figure 6 shows the light model of Section 4.2.6 and its per-lamp metrics on an illustrative street with five lamps at a mounting height of 9 m.

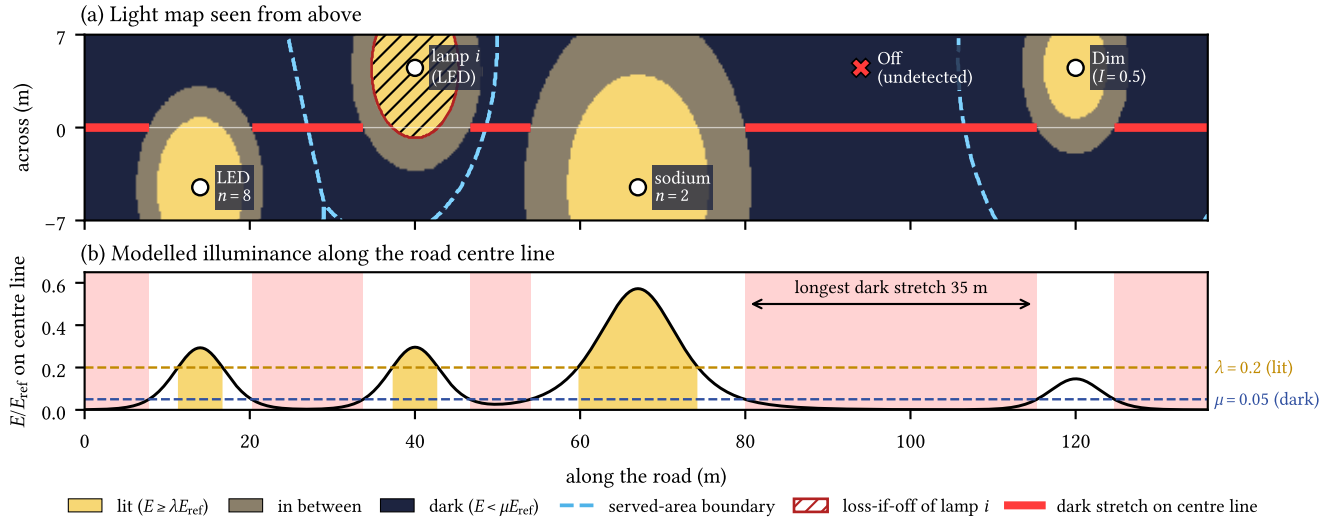


Figure 6: Light-map metrics on an illustrative street. (a) Top-down light map with lit, in-between and dark road, the served-area boundaries, the loss-if-off area of lamp i and the dark stretches on the centre line. (b) Modeled illuminance along the centre line with the lit (λ) and dark (μ) thresholds; the undetected Off lamp leaves the longest dark stretch.

C Annotated Scenes

Figure 7 shows annotated frames from six scene types in the principal dataset.



Figure 7: Annotated frames from six scene types: lamp-head, confounder and ignore boxes, and road, building, confounding-light and ignored pixels.

D Detector Failure Cases

Figure 8 shows typical detector errors on the held-out test recording, at the deployed confidence threshold of 0.15.

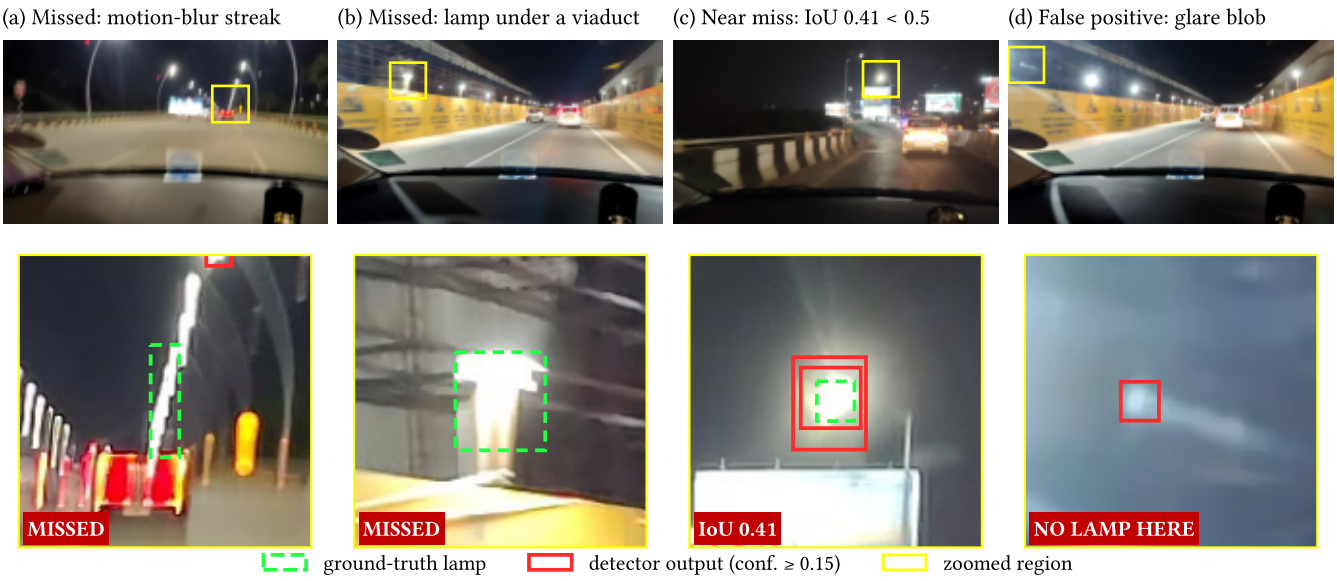


Figure 8: Detector failure cases on the held-out test recording. Top: full frames, with the zoomed region in yellow. Bottom: ground-truth lamps (green, dashed) and detector outputs (red). (a) A lamp smeared into a motion-blur streak is missed. (b) A lamp mounted under a viaduct is missed. (c) The predicted boxes include the glow and overlap the lamp with IoU $0.41 < 0.5$. (d) A glare blob is detected as a lamp.